

Assignment 1

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Question 1: Analyzing the diamonds Dataset

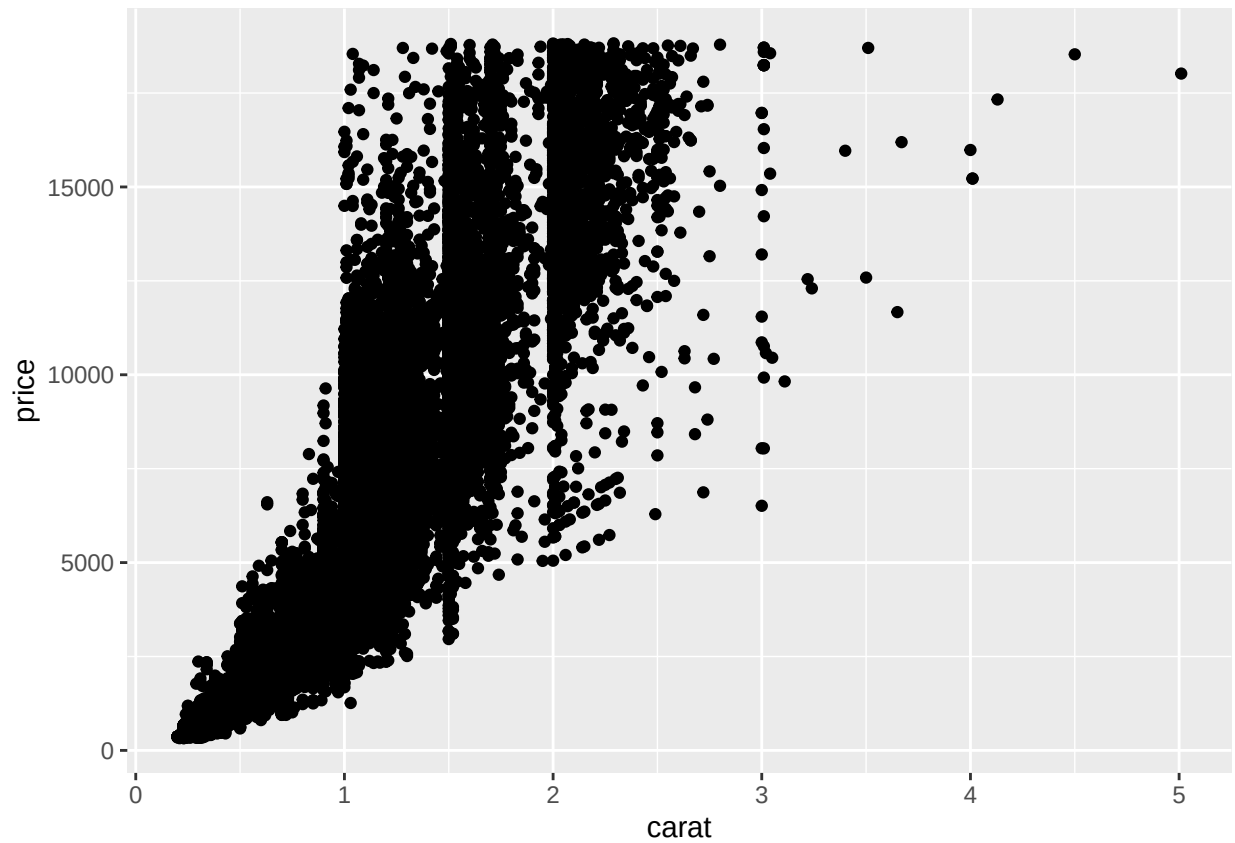
(a) Load the diamonds dataset and display the first few rows.

```
#install.packages("ggplot2")  
#Load the ggplot2 library  
library(ggplot2)  
#Load the diamonds dataset  
data(diamonds)  
#Display first few rows of the dataset  
head(diamonds)
```

```
## # A tibble: 6 x 10  
##   carat cut      color clarity depth table price      x      y      z  
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>  
## 1  0.23 Ideal     E     SI2     61.5    55   326   3.95   3.98   2.43  
## 2  0.21 Premium  E     SI1     59.8    61   326   3.89   3.84   2.31  
## 3  0.23 Good     E     VS1     56.9    65   327   4.05   4.07   2.31  
## 4  0.29 Premium  I     VS2     62.4    58   334   4.2    4.23   2.63  
## 5  0.31 Good     J     SI2     63.3    58   335   4.34   4.35   2.75  
## 6  0.24 Very Good J     VVS2     62.8    57   336   3.94   3.96   2.48
```

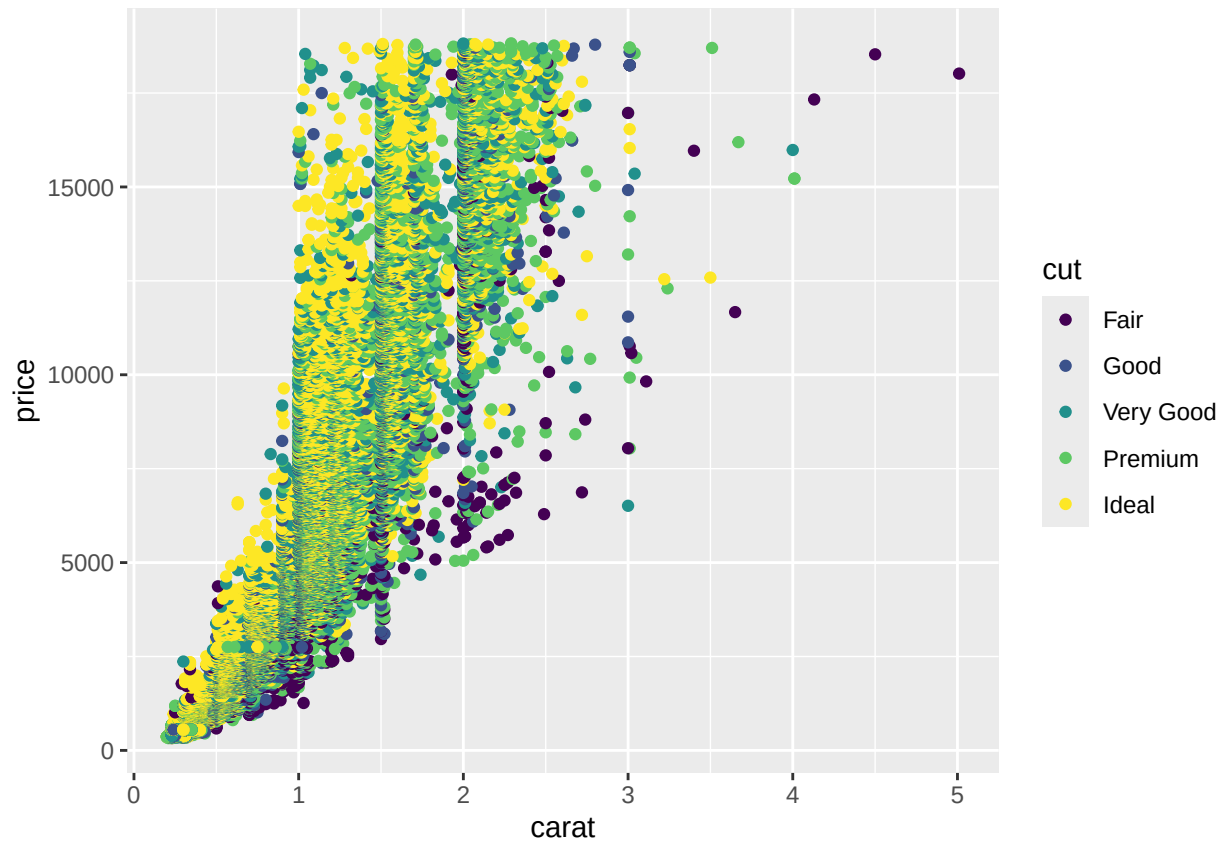
(b) Create a scatter plot of price vs carat.

```
# Create a scatter plot of price vs carat  
ggplot(data = diamonds, aes(x = carat, y = price)) +  
  geom_point()
```



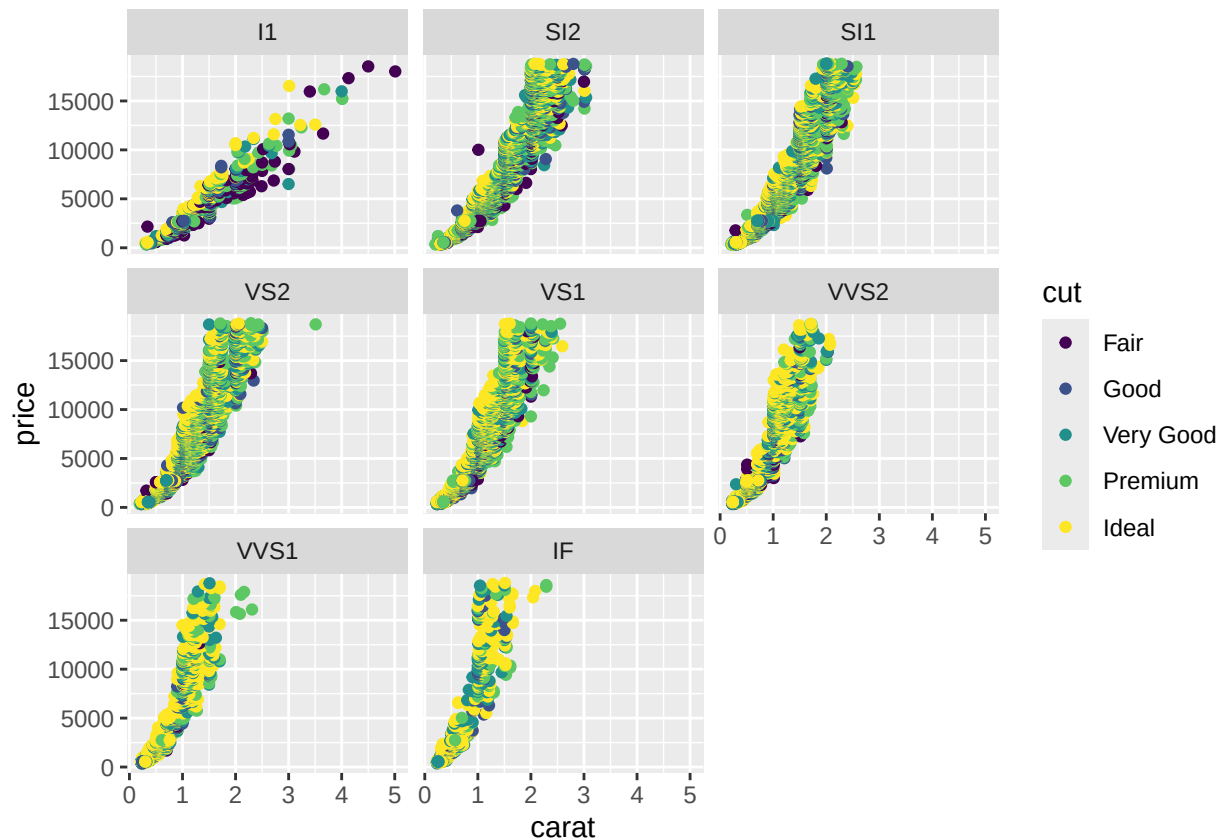
(c) Color the points by the cut of the diamond.

```
# Scatter plot of price vs carat, color points by cut  
ggplot(data = diamonds, aes(x = carat, y = price, color = cut)) +  
  geom_point()
```



(d) Facet the plot by the clarity of the diamond.

```
# Scatter plot of price vs carat, color points by cut, facet by clarity  
ggplot(data = diamonds, aes(x = carat, y = price, color = cut)) +  
  geom_point() +  
  facet_wrap(~ clarity)
```



(e) Add smooth lines to the scatter plot to show trends.

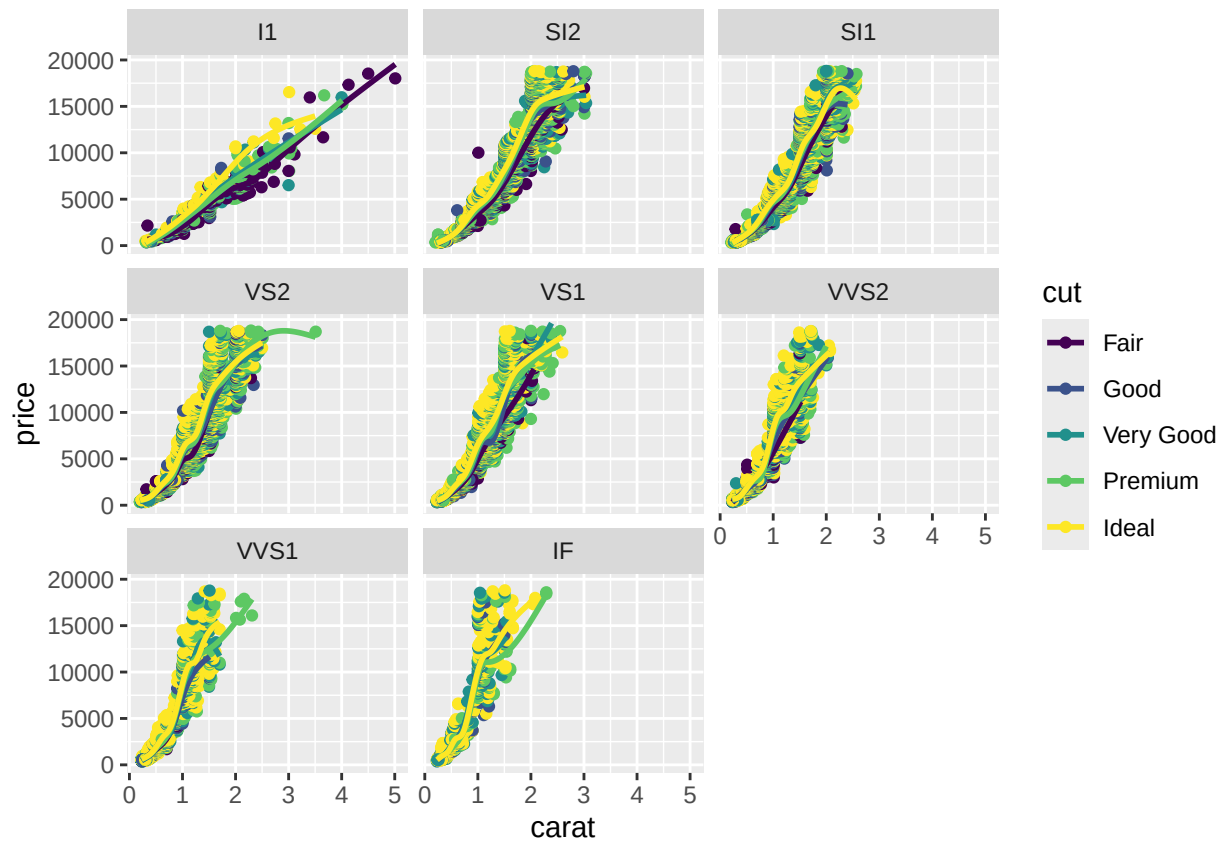
```
# Scatter plot of price vs carat, color points by cut, facet by clarity, with smooth lines
ggplot(data = diamonds, aes(x = carat, y = price, color = cut)) +
  geom_point() +
  geom_smooth(se = FALSE) +
  facet_wrap(~ clarity)
```

```
## 'geom_smooth()' using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

```
## Warning: Failed to fit group 1.
```

```
## Caused by error in 'smooth.construct.cr.smooth.spec()':
```

```
## ! x has insufficient unique values to support 10 knots: reduce k.
```



Question 2:

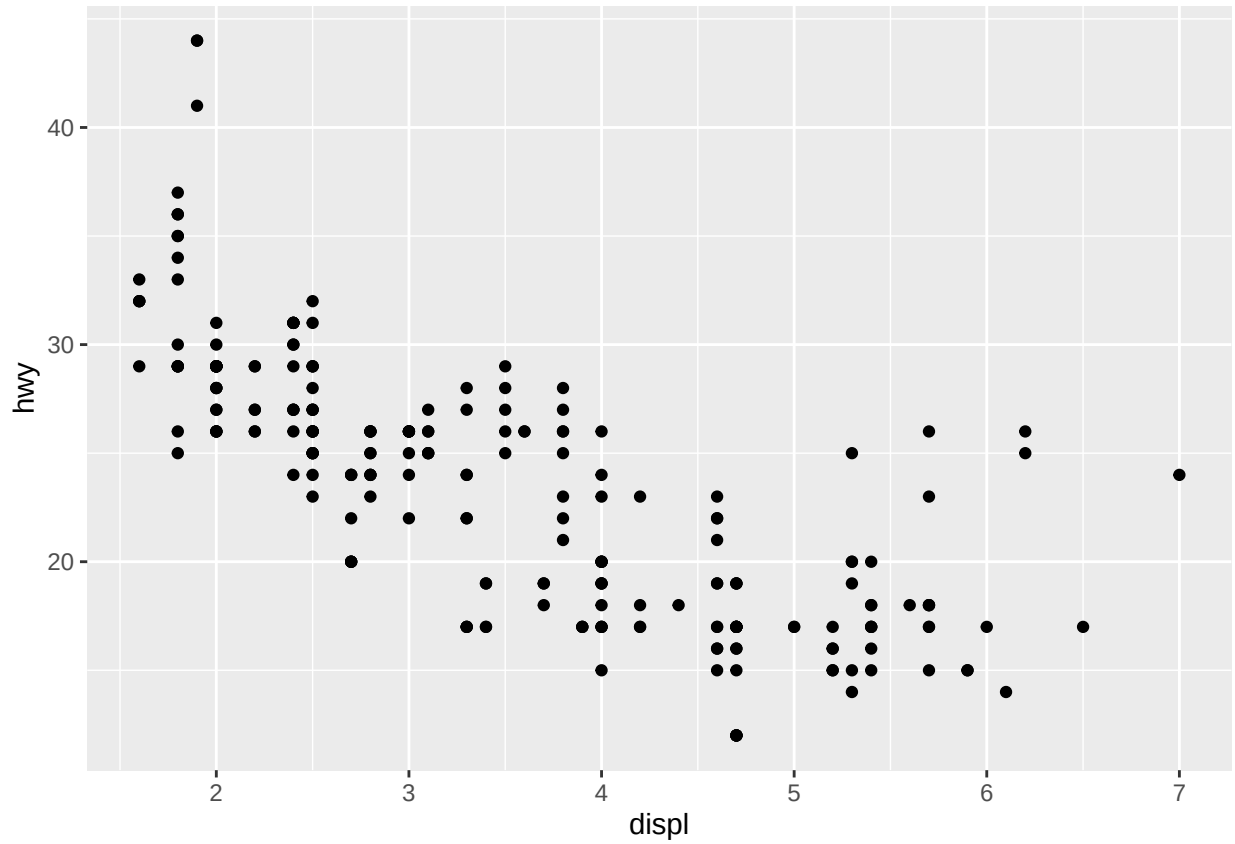
(a) Load the mpg data set and display the first few rows.

```
# Load the mpg data set
data(mpg)
# Display first few rows of the data set
head(mpg)
```

```
## # A tibble: 6 x 11
##   manufacturer model displ  year   cyl trans      drv   cty   hwy fl   class
##   <chr>          <chr> <dbl> <int> <int> <chr>    <chr> <int> <int> <chr> <chr>
## 1 audi          a4      1.8  1999     4 auto(l5)  f      18    29 p   compa~
## 2 audi          a4      1.8  1999     4 manual(m5) f      21    29 p   compa~
## 3 audi          a4      2    2008     4 manual(m6) f      20    31 p   compa~
## 4 audi          a4      2    2008     4 auto(av)   f      21    30 p   compa~
## 5 audi          a4      2.8  1999     6 auto(l5)  f      16    26 p   compa~
## 6 audi          a4      2.8  1999     6 manual(m5) f      18    26 p   compa~
```

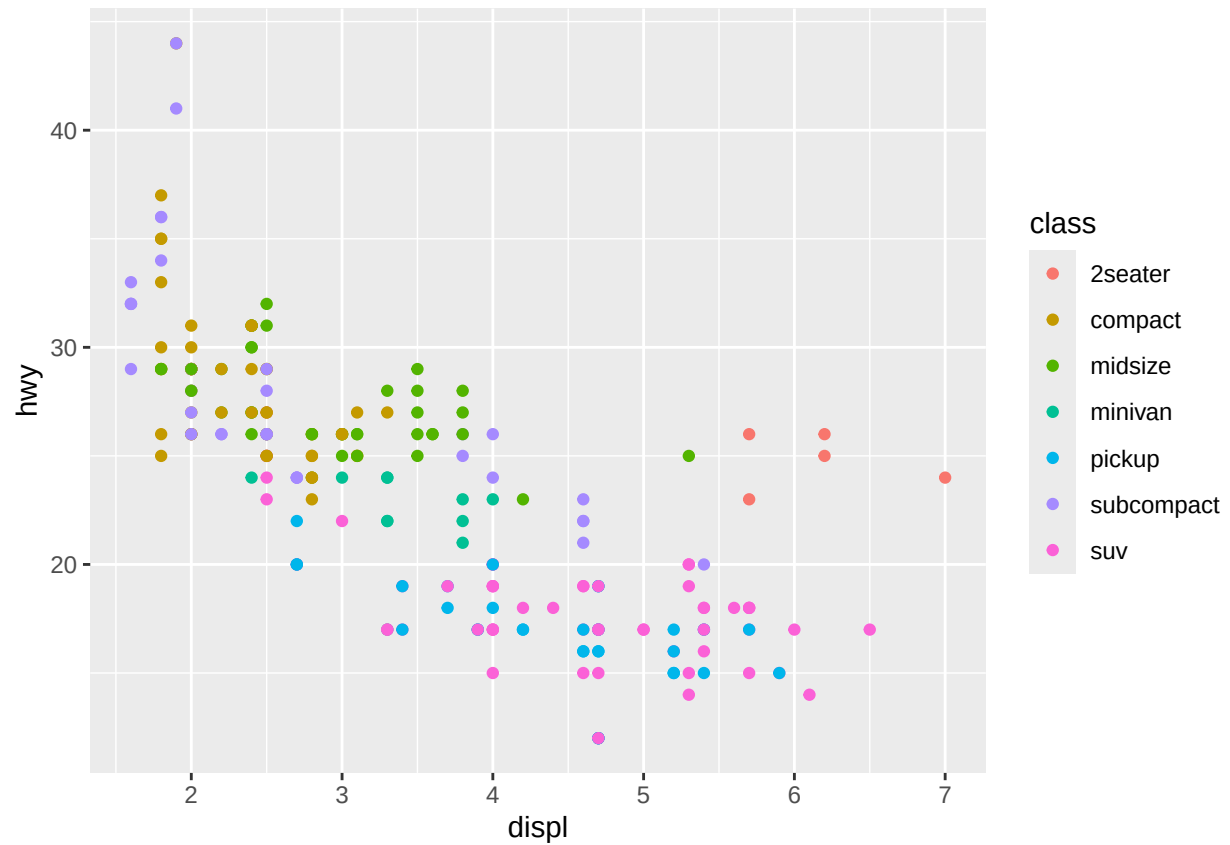
(b) Create a scatter plot of hwy (highway miles per gallon) vs displ (engine displacement)

```
# Scatter plot of hwy vs displ
ggplot(data = mpg, aes(x = displ, y = hwy)) +
  geom_point()
```



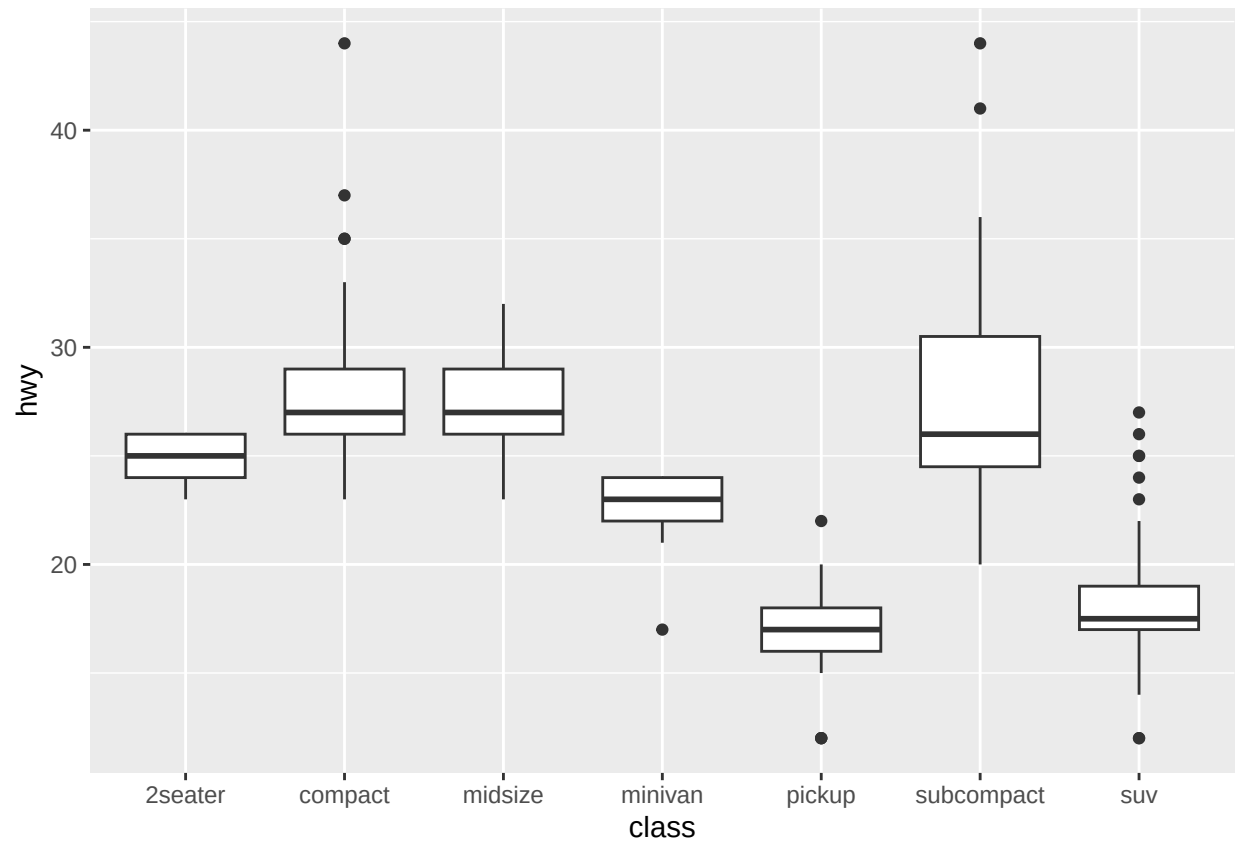
(c) Color the points by the class of the vehicle

```
# Scatter plot of hwy vs displ, color points by class
ggplot(data = mpg, aes(x = displ, y = hwy, color = class)) +
  geom_point()
```



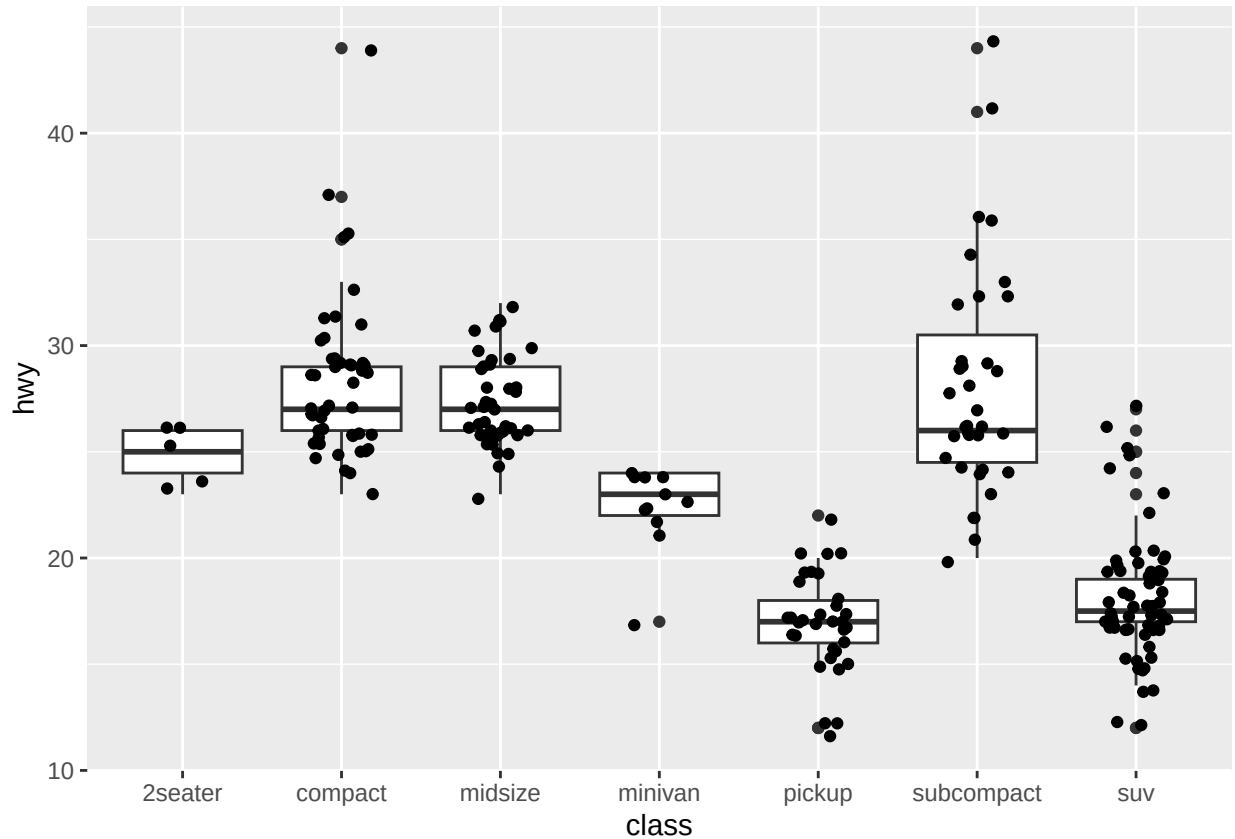
(d) Create a boxplot of hwy grouped by class.

```
# Boxplot of hwy grouped by class
ggplot(data = mpg, aes(x = class, y = hwy)) +
  geom_boxplot()
```



(e) Add jitter to the boxplot to show individual data points.

```
# Boxplot of hwy grouped by class, with jittered individual points  
ggplot(data = mpg, aes(x = class, y = hwy)) +  
  geom_boxplot() +  
  geom_jitter(width = 0.2)
```

Question 3: Rental Car Company

(a) Define the response variable, the factor, and the treatment.

- **Response variable:** length of the rental period.
- **Factor:** Car type
- **Treatments:** The four different car types (Subcompact, Compact, Midsize, Full Size) being applied

(b) What are the experimental units?

- **Experimental units:** The individual rental contracts.

(c) State the hypotheses needed to conduct a one-way ANOVA. Perform a one-way ANOVA procedure on the data above. Is there sufficient evidence at the 5% level of significance to indicate that there is a difference in the mean rental length under the different car type? Use the p-value approach.

- **H0:** The mean rental length is the same for all car types.
- **H1:** At least one car type has a different mean rental length.

```

# Enter the data from the chart in question 3 by creating 2 vectors for rental length and car type
rental_length <- c(
  # Subcompact
  2, 1, 2, 1, 2, 1, 1, 1, 3, 2,
  # Compact
  12, 5, 2, 7, 7, 5, 7, 8, 12, 10,
  # Midsize
  8, 3, 3, 5, 1, 3, 1, 1, 3, 4,
  # Full Size
  4, 1, 9, 3, 3, 4, 9, 2, 2, 6
)

car_type <- factor(rep(c("Subcompact", "Compact", "Midsize", "Full Size"), each = 10))

# Create a data frame
rental_data <- data.frame(rental_length, car_type)

# Perform One-Way ANOVA
anova_result <- aov(rental_length ~ car_type, data = rental_data)

# Display the ANOVA table
summary(anova_result)

```

```

##           Df Sum Sq Mean Sq F value    Pr(>F)
## car_type    3  186.5   62.17   10.73 3.51e-05 ***
## Residuals   36   208.6    5.79
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

- Based on the p-value (3.51e-05), we reject the null hypothesis at the 5% significance level because the p-value is much smaller than 0.05, which shows there could be differences between car types.
- There is sufficient evidence to suggest that mean rental length differs between car types.

(d) Use a 95% Tukey simultaneous confidence interval. Discuss the nature of any differences in the mean rental length under the different car type. Is there a particular type that is a clear winner when it comes to mean rental length?

```

# run Tukey
tukey_result <- TukeyHSD(anova_result)

# Display Tukey test result
tukey_result

```

```

##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = rental_length ~ car_type, data = rental_data)
##
## $car_type
##              diff              lwr              upr              p adj

```

```
## Full Size-Compact      -3.2 -6.099305 -0.3006951 0.0257558
## Midsize-Compact       -4.3 -7.199305 -1.4006951 0.0016726
## Subcompact-Compact    -5.9 -8.799305 -3.0006951 0.0000198
## Midsize-Full Size     -1.1 -3.999305  1.7993049 0.7378653
## Subcompact-Full Size  -2.7 -5.599305  0.1993049 0.0755939
## Subcompact-Midsize    -1.6 -4.499305  1.2993049 0.4560529
```

- A Tukey HSD test was performed to compare mean rental lengths between car types at a 95% confidence level.
- The following pairs of car types showed significant differences in mean rental length ($p \text{ adj} < 0.05$):
 - Compact > Full Size
 - Compact > Midsize
 - Compact > Subcompact
- Since Compact is greater than all 3 other car types, Compact is the clear winner because it has the highest mean rental length.

Question 4: Chemist

(a) Define the response variable, treatment factor, block factor.

- **Response variable:** Tensile Strength of cloth
- **Treatment factor:** Chemical agent used (Chemical 1, Chemical 2, Chemical 3, Chemical 4).
- **Block factor:** Bolt of cloth (Bolt 1, Bolt 2, Bolt 3, Bolt 4, Bolt 5).

(b) Give the ANOVA Table.

```
TensileStrength ~ Chemical + Block
```

```
## TensileStrength ~ Chemical + Block
```

```
# Enter the data from the chart in question 4 by creating 3 vectors for tensile Strength, Chemical, Bolt
```

```
# Response variable: Tensile strength
```

```
TensileStrength <- c(
  # Chemical 1
  73, 68, 74, 71, 67,
  # Chemical 2
  73, 67, 75, 72, 70,
  # Chemical 3
  75, 68, 78, 73, 68,
  # Chemical 4
  73, 71, 75, 75, 69
)
```

```
# Treatment factor: Chemical agent (1 to 4)
```

```
Chemical <- factor(rep(c("1", "2", "3", "4"), each = 5))
```

```

# Block factor: Bolt of cloth (1 to 5), repeated for each chemical
Block <- factor(rep(c("1", "2", "3", "4", "5"), times = 4))

# Create data frame
cloth_data <- data.frame(TensileStrength, Chemical, Block)

# Perform ANOVA
anova_cloth <- aov(TensileStrength ~ Chemical + Block, data = cloth_data)

# Display ANOVA table
summary(anova_cloth)

```

```

##           Df Sum Sq Mean Sq F value    Pr(>F)
## Chemical    3  12.95    4.32    2.376    0.121
## Block       4 157.00   39.25   21.606 2.06e-05 ***
## Residuals   12   21.80    1.82
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(c) Is there sufficient evidence at $\alpha = 0.05$ to indicate that there is a difference in the mean tensile strengths from each chemical. Use a critical value approach, and make sure to define each mean in your hypotheses.

- **H0:** The mean tensile strength is the same for all Chemical agents.
- **H1:** At least one Chemical agent has a different mean tensile strength.

```

# Calculate critical value for Chemical effect (alpha = 0.05)
qf(0.95, df1 = 3, df2 = 12)

```

```
## [1] 3.490295
```

- The calculated critical value is 3.49.
- Since the F value for Chemical (2.376) is less than the critical value (3.49), we fail to reject the null hypothesis.
- At the 5% significance level, there is not enough evidence to conclude that mean tensile strength differs across Chemical agents.

(d) Is there sufficient evidence at $\alpha = 0.05$ to indicate that there is a difference in the mean tensile strength for each bolt? Use a critical value approach, and make sure to define each mean in your hypotheses.

- **H0:** The mean tensile strength is the same for all Blocks of cloth.
- **H1:** At least one Block of cloth has a different mean tensile strength.

```

# Calculate critical value for Block effect (alpha = 0.05)
qf(0.95, df1 = 4, df2 = 12)

```

```
## [1] 3.259167
```

- The calculated critical value is 3.26.
- Since the F value for Block (21.606) is greater than the critical value (3.26), we reject the null hypothesis.
- At the 5% significance level, there is sufficient evidence to conclude that mean tensile strength differs across Blocks of cloth.

(e) Compute a 95% simultaneous Tukey confidence interval to discuss the nature of any differences in the mean tensile strength.

```
# Perform Tukey HSD test for Chemical effect
TukeyHSD(anova_cloth)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = TensileStrength ~ Chemical + Block, data = cloth_data)
##
## $Chemical
##      diff      lwr      upr      p adj
## 2-1  0.8 -1.7308322  3.330832 0.7852734
## 3-1  1.8 -0.7308322  4.330832 0.2042593
## 4-1  2.0 -0.5308322  4.530832 0.1417326
## 3-2  1.0 -1.5308322  3.530832 0.6540138
## 4-2  1.2 -1.3308322  3.730832 0.5182726
## 4-3  0.2 -2.3308322  2.730832 0.9952030
##
## $Block
##      diff      lwr      upr      p adj
## 2-1 -5.00 -8.037831 -1.9621691 0.0015656
## 3-1  2.00 -1.037831  5.0378309 0.2814173
## 4-1 -0.75 -3.787831  2.2878309 0.9295872
## 5-1 -5.00 -8.037831 -1.9621691 0.0015656
## 3-2  7.00  3.962169 10.0378309 0.0000717
## 4-2  4.25  1.212169  7.2878309 0.0056966
## 5-2  0.00 -3.037831  3.0378309 1.0000000
## 4-3 -2.75 -5.787831  0.2878309 0.0830636
## 5-3 -7.00 -10.037831 -3.9621691 0.0000717
## 5-4 -4.25 -7.287831 -1.2121691 0.0056966
```

- Based on the Tukey HSD test, there is no significant difference in mean tensile strength between any pairs of Chemical agents.
- All 95% confidence intervals contain 0, so we cannot conclude that any Chemical is better than the others.

Question 5

(a) Define the treatments

Each treatment corresponds to a specific pair of temperature and pressure values:

- (150, 200), (150, 215), (150, 230)
- (160, 200), (160, 215), (160, 230)
- (170, 200), (170, 215), (170, 230)

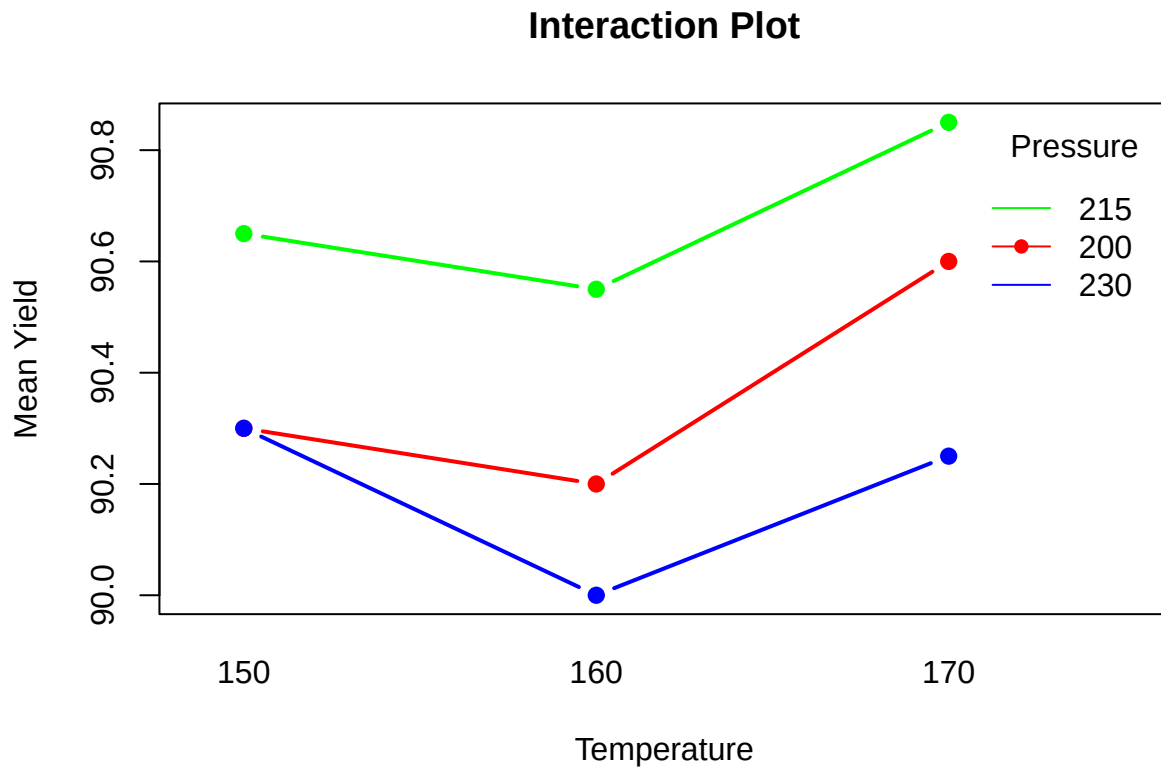
(b) Perform a graphical analysis to check for interaction between the temperature and pressure.

```
# Create the factors manually
Temperature <- factor(rep(c(150, 160, 170), each = 6))
Pressure <- factor(rep(rep(c(200, 215, 230), each = 2), times = 3))

# Enter the yield values in order shown above
Yield <- c(
  90.4, 90.2, 90.7, 90.6, 90.2, 90.4,
  90.1, 90.3, 90.5, 90.6, 89.9, 90.1,
  90.5, 90.7, 90.8, 90.9, 90.4, 90.1
)

# Combine into a data frame
data_q5 <- data.frame(
  Temperature,
  Pressure,
  Yield
)

interaction.plot(
  x.factor = data_q5$Temperature,
  trace.factor = data_q5$Pressure,
  response = data_q5$Yield,
  fun = mean,
  type = "b",
  col = c("red", "green", "blue"),
  pch = 16,
  lty = 1,
  lwd = 2,
  cex = 1.2,
  xlab = "Temperature",
  ylab = "Mean Yield",
  trace.label = "Pressure",
  main = "Interaction Plot"
)
```



(c) Test for interaction with $\alpha = 0.05$. State your hypotheses

H0: There is no interaction between temperature and pressure on yield. H1: There is an interaction between temperature and pressure on yield.

```
# Define the data
Yield <- c(
  90.4, 90.2, 90.7, 90.6, 90.2, 90.4,
  90.1, 90.3, 90.5, 90.6, 89.9, 90.1,
  90.5, 90.7, 90.8, 90.9, 90.4, 90.1
)

Temperature <- factor(rep(c(150, 160, 170), each = 6))
Pressure <- factor(rep(rep(c(200, 215, 230), each = 2), 3))

# Create a data frame
data_q5 <- data.frame(Yield, Temperature, Pressure)

# Perform two-way ANOVA with interaction
anova_q5 <- aov(Yield ~ Temperature * Pressure, data = data_q5)

# Display ANOVA table
summary(anova_q5)
```

##	Df	Sum Sq	Mean Sq	F value	Pr(>F)
----	----	--------	---------	---------	--------

```
## Temperature      2 0.3011  0.1506   8.469 0.008539 **
## Pressure         2 0.7678  0.3839  21.594 0.000367 ***
## Temperature:Pressure 4 0.0689  0.0172   0.969 0.470006
## Residuals        9 0.1600  0.0178
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Since the p-value for the interaction between temperature and pressure is 0.4701, which is greater than 0.05, we fail to reject the null hypothesis.
- At the 5% significance level, there is no significant interaction between temperature and pressure. This suggests that the effect of temperature on yield does not depend on the pressure level, and vice versa.

(d) Test the significance of temperature effects with $\alpha = 0.05$. State your hypotheses.

- H_0 : Temperature has no effect on yield.
- H_1 : Temperature has a significant effect on yield.
- Since the p-value for temperature is 0.0058, which is less than 0.05, we reject the null hypothesis.
- At the 5% significance level, there is significant evidence to suggest that temperature has an effect on yield.

(e) Test the significance of pressure effects with $\alpha = 0.05$. State your hypotheses.

- H_0 : Pressure has no effect on mean yield
- H_1 : Pressure has a significant effect on mean yield
- Since $0.00837 < 0.05$, we reject the null hypothesis.
- At the 5% significance level, there is sufficient evidence to conclude that pressure has a significant effect on mean yield. This means that the different pressure levels (200, 215, 230) do not produce the same yield, on average.

(f) Make a pairwise comparison using Tukey simultaneous 95% confidence interval

```
# Perform Tukey pairwise comparison
TukeyHSD(anova_q5)

##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = Yield ~ Temperature * Pressure, data = data_q5)
##
## $Temperature
##           diff           lwr           upr           p adj
## 160-150 -0.1666667 -0.38159536 0.04826203 0.1313152
```



```

## 170-150  0.1500000 -0.06492869 0.36492869 0.1809078
## 170-160  0.3166667  0.10173797 0.53159536 0.0066518
##
## $Pressure
##           diff           lwr           upr           p adj
## 215-200  0.3166667  0.1017380  0.53159536 0.0066518
## 230-200 -0.1833333 -0.3982620  0.03159536 0.0944905
## 230-215 -0.5000000 -0.7149287 -0.28507131 0.0002951
##
## $'Temperature:Pressure'
##           diff           lwr           upr           p adj
## 160:200-150:200 -1.000000e-01 -0.62747453  0.42747453 0.9959112
## 170:200-150:200  3.000000e-01 -0.22747453  0.82747453 0.4489114
## 150:215-150:200  3.500000e-01 -0.17747453  0.87747453 0.2916707
## 160:215-150:200  2.500000e-01 -0.27747453  0.77747453 0.6429825
## 170:215-150:200  5.500000e-01  0.02252547  1.07747453 0.0397898
## 150:230-150:200 -8.526513e-14 -0.52747453  0.52747453 1.0000000
## 160:230-150:200 -3.000000e-01 -0.82747453  0.22747453 0.4489114
## 170:230-150:200 -5.000000e-02 -0.57747453  0.47747453 0.9999713
## 170:200-160:200  4.000000e-01 -0.12747453  0.92747453 0.1812587
## 150:215-160:200  4.500000e-01 -0.07747453  0.97747453 0.1099996
## 160:215-160:200  3.500000e-01 -0.17747453  0.87747453 0.2916707
## 170:215-160:200  6.500000e-01  0.12252547  1.17747453 0.0147551
## 150:230-160:200  1.000000e-01 -0.42747453  0.62747453 0.9959112
## 160:230-160:200 -2.000000e-01 -0.72747453  0.32747453 0.8316762
## 170:230-160:200  5.000000e-02 -0.47747453  0.57747453 0.9999713
## 150:215-170:200  5.000000e-02 -0.47747453  0.57747453 0.9999713
## 160:215-170:200 -5.000000e-02 -0.57747453  0.47747453 0.9999713
## 170:215-170:200  2.500000e-01 -0.27747453  0.77747453 0.6429825
## 150:230-170:200 -3.000000e-01 -0.82747453  0.22747453 0.4489114
## 160:230-170:200 -6.000000e-01 -1.12747453 -0.07252547 0.0241003
## 170:230-170:200 -3.500000e-01 -0.87747453  0.17747453 0.2916707
## 160:215-150:215 -1.000000e-01 -0.62747453  0.42747453 0.9959112
## 170:215-150:215  2.000000e-01 -0.32747453  0.72747453 0.8316762
## 150:230-150:215 -3.500000e-01 -0.87747453  0.17747453 0.2916707
## 160:230-150:215 -6.500000e-01 -1.17747453 -0.12252547 0.0147551
## 170:230-150:215 -4.000000e-01 -0.92747453  0.12747453 0.1812587
## 170:215-160:215  3.000000e-01 -0.22747453  0.82747453 0.4489114
## 150:230-160:215 -2.500000e-01 -0.77747453  0.27747453 0.6429825
## 160:230-160:215 -5.500000e-01 -1.07747453 -0.02252547 0.0397898
## 170:230-160:215 -3.000000e-01 -0.82747453  0.22747453 0.4489114
## 150:230-170:215 -5.500000e-01 -1.07747453 -0.02252547 0.0397898
## 160:230-170:215 -8.500000e-01 -1.37747453 -0.32252547 0.0023855
## 170:230-170:215 -6.000000e-01 -1.12747453 -0.07252547 0.0241003
## 160:230-150:230 -3.000000e-01 -0.82747453  0.22747453 0.4489114
## 170:230-150:230 -5.000000e-02 -0.57747453  0.47747453 0.9999713
## 170:230-160:230  2.500000e-01 -0.27747453  0.77747453 0.6429825

```

- for temperature, There is a significant difference between 170 and both 150 and 160 ($p < 0.05$). No significant difference was found between 150 and 160.
- for pressure, levels 215 and 230 both differ significantly from 200 ($p < 0.05$), but there is no significant difference between 215 and 230.

(g) Which temperature and pressure maximizes yied? Does your answers depend on each other? Why?

```
# Compute mean yield by Temperature and Pressure  
aggregate(Yield ~ Temperature + Pressure, data = data_q5, FUN = mean)
```

```
##   Temperature Pressure Yield  
## 1         150      200 90.30  
## 2         160      200 90.20  
## 3         170      200 90.60  
## 4         150      215 90.65  
## 5         160      215 90.55  
## 6         170      215 90.85  
## 7         150      230 90.30  
## 8         160      230 90.00  
## 9         170      230 90.25
```

- The highest mean yield is **90.85**, which happens when the **temperature is 170°C** and **pressure is 215**.
- So if we're looking to maximize yield, the best combo is:
 - **Temperature: 170**
 - **Pressure: 215**
- Even though the interaction term wasn't significant ($p = 0.47$), both temperature and pressure still have individual effects on yield.
- That said, since the highest yield only shows up when both of these are at specific levels (170 and 215), it's pretty clear that **the best result depends on both factors working together**, not just one or the other.